Neural Unbalanced Optimal Transport via Cycle-Consistent Semi-Couplings

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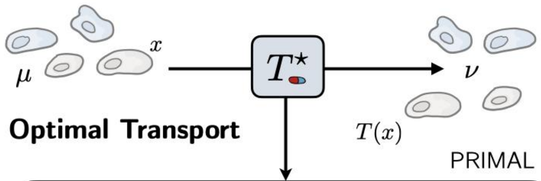


Optimal Transport

PRIMAL

$$T^\star \leftarrow \inf_{T:T_\sharp\mu\approx\nu} \int_{\mathbb{R}^d} \|x - T(x)\|_2^2 d\mu(x)$$

.. reconstruct alignment between unperturbed and perturbed cell states via a **mapping** that minimizes the overall **displacement cost** when connecting each cell x to its most likely perturbed state $T(x)$

$T^\star$: Monge map

.. **Brenier theorem** (1987) states that there must exist a convex potential $\psi^\star$ s.t. $T^\star = \nabla\psi^\star$.

$$\psi^\star \leftarrow \arg\sup_{\psi \text{ convex}} \int_{\mathbb{R}^d} \psi^* \mathrm{d}\mu + \int_{\mathbb{R}^d} \psi \mathrm{d}\nu$$

SEMI-DUAL

.. in practice, it's more efficient to parameterize an approximation of

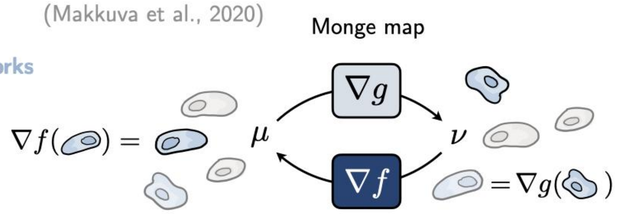
DUAL

$$(g^\star, f^\star) \leftarrow \sup_{(f,g) \text{ convex}} \int_{\mathbb{R}^d} g(x)\mathrm{d}\mu(x) + \int_{\mathbb{R}^d} f(y)\mathrm{d}\nu(y)$$

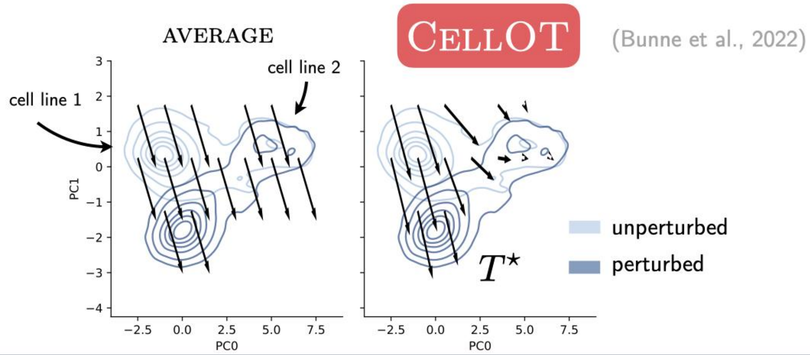
(Makkuva et al., 2020)

parameterize g and f via **input convex neural networks**

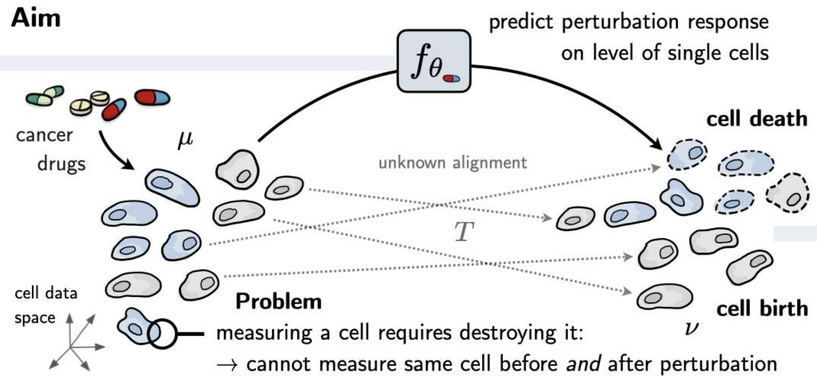
(Amos et al., 2017)



.. CellOT is **state-of-the-art** in predicting perturbation responses of heterogeneous populations on the level of single cells



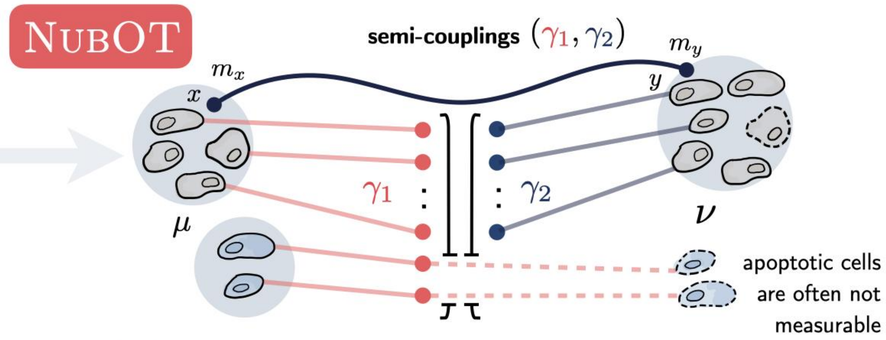
Aim



However, cells can die and proliferate as a response to treatment.

.. solve an **unbalanced** optimal transport problem

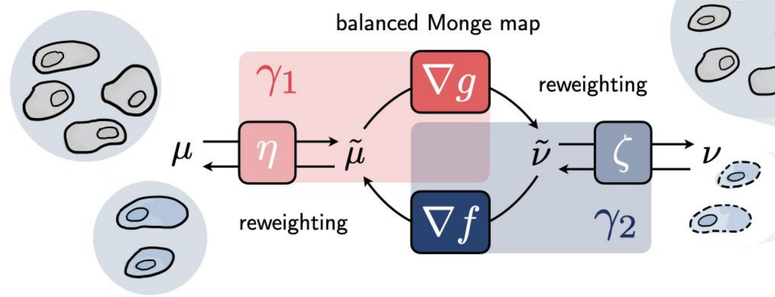
(Chizat et al., 2018)

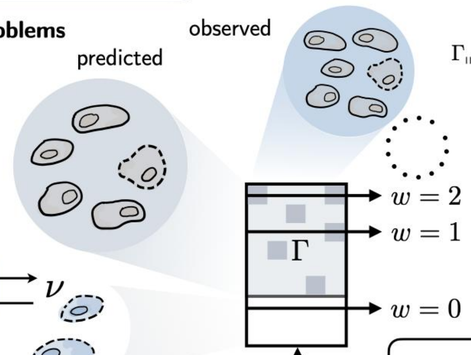


Conceptually break problem into **balanced** and **unbalanced subproblems**

.. learn balanced proxy measures, i.e., $\tilde{\mu} = \eta \cdot \mu$ and $\tilde{\nu} = \zeta \cdot \nu$

.. solve balanced OT problem between proxy measures





$$\Gamma_{\shortparallel}(\mu,\nu) \overset{\text{def.}}{=} \left\{(\gamma_1,\gamma_2) \in (\mathcal{M}_+(\mathcal{X}^2))^2 : (\mathrm{Proj}_1)_\sharp\gamma_1 = \mu, (\mathrm{Proj}_2)_\sharp\gamma_2 = \nu\right\}$$

.. describe how much mass is transported between any pair $x, y \in \mathcal{X}$, while allowing for variation of mass.

predicted $\hat{y} = \nabla g(x)$ observed y

$$\Gamma \leftarrow \mathrm{UBOT}(\hat{\nu}, \nu)$$

discrete unbalanced optimal transport

.. to predict out-of-sample, train continuous version η via

$$e_i \leftarrow \frac{\sum_j \Gamma_{ij}}{\sum_{ij} \Gamma_{ij}} \cdot n$$

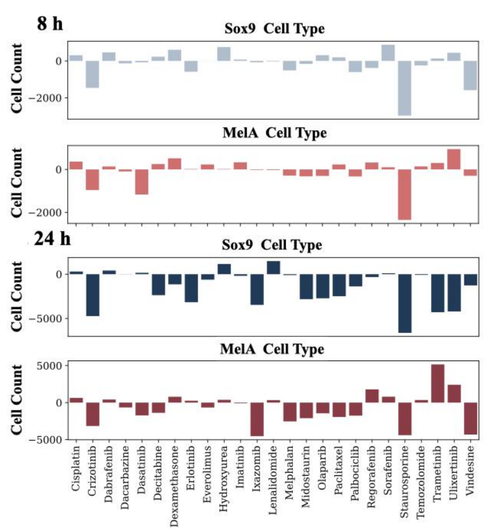
$$\mathrm{loss}_\eta(\theta_\eta) = \mathrm{MSE}(\mathbf{e}, \eta(x))$$

Evaluation

Observed changes in cell counts per cell line

.. give insights into treatment-induced death and proliferation

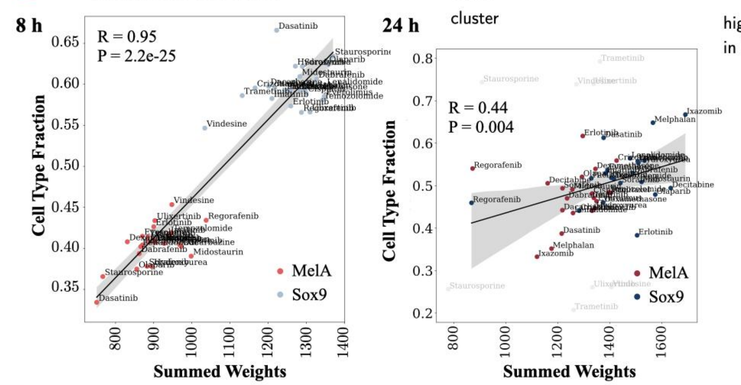
.. and provide an assessment of the weights learned by NubOT

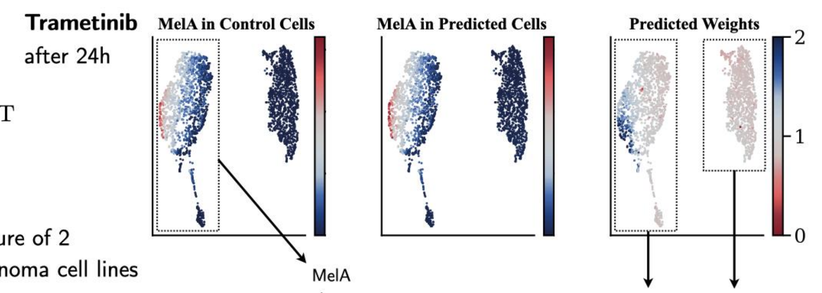


Sox9

MelA

mixture of 2 melanoma cell lines





*equal contribution